Lab 3 — Final Technical Report

Course: EEC 174AY

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Executive summary

This lab investigated convolutional networks from small handwritten-digit classification to transfer learning and fine-grained vehicle recognition. Phase 1 examined hyperparameter sensitivity on MNIST; Phase 2 compared augmentation, full fine-tuning, and frozen ResNet18 configurations on CIFAR10; Phase 3 trained an ImageNet-pretrained ResNet50 to classify 196 Stanford Cars models. The best reported results are 99.19% MNIST test accuracy, 84.05% CIFAR10 test accuracy, and 83.87% Stanford Cars validation accuracy. The datasets, splits, and metrics differ, so these percentages should not be treated as a head-to-head model ranking.

Phase	Dataset / task	Best recorded score	Evaluation split
1	MNIST / 10 handwritten digits	99.19%	Test
2	CIFAR10 / 10 object classes	84.05%	Test
3	Stanford Cars / 196 models	83.87%	Validation

1. Phase 1 — CNN on MNIST

Objective. Implement and compare convolutional models on 28×28 grayscale handwritten digit images, then investigate learning rate, optimizer, initialization, activation function, batch size, network architecture, and batch normalization. Training and learning-curve functions were completed in the provided notebook.

1.1 Hyperparameter experiments

Experiment	Setting	Recorded final test accuracy
Learning rate	0.0003 / 0.002 / 0.05	97.94% / 97.89% / 94.76%
Optimizer	SGD / SGD momentum 0.5 / 0.9	92.30% / 92.95% / 96.91%
Initialization	Zero / all ones / normal	11.35% / 97.64% / 97.99%
Activation	tanh	94.42%
Batch size	8 / 64 / 128	98.03% / 97.88% / 97.96%
Architecture	Net2 / Net5 / FCN	98.12% / 99.01% / 97.52%
Batch norm	Net5 + batch normalization	99.19%

Interpretation. The highest learning rate (0.05) produced weaker accuracy than 0.0003 or 0.002 in this experiment. Momentum improved SGD markedly, especially at 0.9. All-zero initialization largely prevented useful learning, consistent with symmetry across units. The tanh experiment scored below the relevant ReLU comparison. Smaller batches

provided little final accuracy benefit relative to their greater per-epoch cost. Deeper convolutional features were beneficial: Net5 outperformed both Net2 and the fully connected baseline, and batch normalization raised the recorded Net5 result from 99.01% to 99.19%. These are experiment-specific outcomes, not universal claims about every training setup.

2. Phase 2 — CNN on CIFAR10

Objective. Classify 32×32 color images in ten categories using an ImageNet-pretrained ResNet18. Implement label-preserving augmentation (including a custom random rotation), compare full-network fine-tuning with and without augmentation, and evaluate increasingly unfrozen network portions.

2.1 Transfer-learning comparison

ResNet18 configuration	Recorded test accuracy
Full fine-tuning, no augmentation	81.96%
Full fine-tuning, augmentation	84.05%
Frozen pretrained backbone; FC layer only	45.94%
Layer 4 + FC trainable	69.00%
Layers 3–4 + FC trainable	77.20%

Interpretation. Data augmentation increased test accuracy by 2.09 percentage points under the tested settings (84.05% − 81.96%). Freezing most of the pretrained backbone reduced trainable parameters but substantially constrained adaptation to CIFAR10. Unfreezing progressively more residual blocks improved performance from 45.94% to 77.20%, while full fine-tuning performed best. The notebook includes a confusion matrix and class-level classification report; any discussion of specific confused class pairs should be grounded in those saved outputs.

3. Phase 3 — Stanford Cars vehicle classification

Objective. Build a 196-class fine-grained vehicle classifier using a custom dataset class and a pretrained convolutional model, then save a usable checkpoint for TA inference on separate hold-out images.

3.1 Data, splits, and custom loader

The downloaded Stanford Cars dataset is arranged in folders named for each vehicle-model class, with 8,144 images in the provided training portion and 8,041 images in a separate test portion. The custom CarDataSet inherits from torch.utils.data.Dataset and implements __init__(), __len__(), __getitem__(), and show_imgs(). It enumerates image files and associates each directory name with a consistent class index. The visualization method displays five examples from distinct classes. A fixed-seed 85/15 split of the supplied training portion produced 6,923 training and 1,221 validation examples. The 8,041-image test portion was not used to produce the reported 83.87% validation result.

3.2 Preprocessing and architecture

Training transforms included random resized crops and horizontal flips; evaluation used a deterministic resize to 256 pixels and center crop to 224×224. ImageNet mean and standard deviation were applied through the transform pipeline. ResNet50 pretrained on ImageNet provided reusable image features, while its last fully connected layer was replaced by a 196-unit classification head. The model was fine-tuned on the vehicle data rather than trained from random weights.

Training parameter	Value
Architecture	ImageNet-pretrained ResNet50
Classes	196

Input size	224 × 224
Batch size	32
Epochs	12
Optimizer	AdamW
Initial learning rate	0.0001
Weight decay	0.0001
Loss	CrossEntropyLoss, label smoothing 0.1
Schedule	CosineAnnealingLR
Checkpoint selection	Highest validation accuracy

3.3 Recorded training results

Metric	Recorded value
Final training accuracy	86.38%
Final validation accuracy	83.87%
Final training loss	1.5188
Final validation loss	1.4723
Best validation accuracy	83.87% (epoch 12)

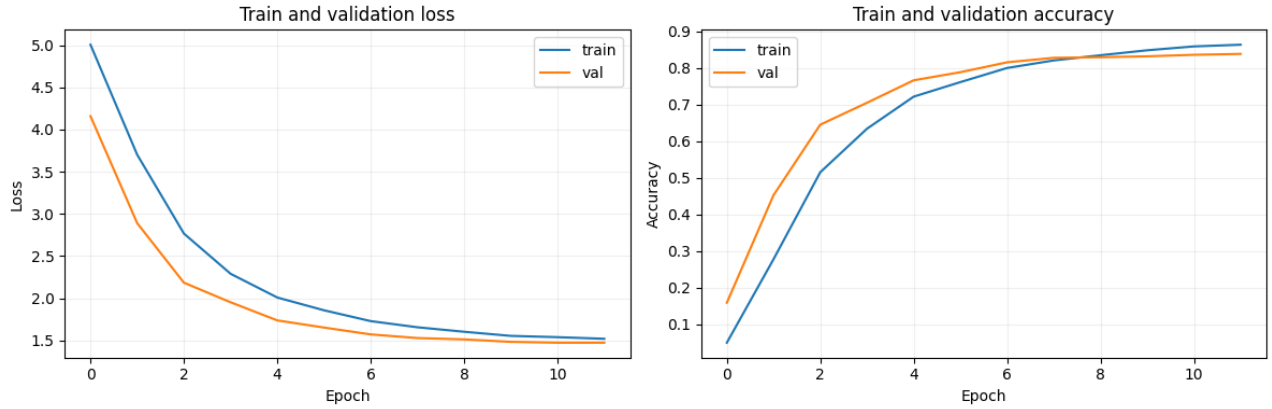


Figure 1. Phase 3 training and validation loss and accuracy curves, extracted from the executed notebook.

Interpretation. Validation accuracy rose from 15.89% in epoch 1 to 83.87% in epoch 12, indicating successful transfer learning. The final train–validation accuracy gap was approximately 2.51 percentage points, which by itself does not indicate severe overfitting. Training and validation losses reflect the label-smoothed objective and should be interpreted alongside accuracy. A stronger generalization assessment requires evaluating the saved checkpoint on truly held-out images without further model selection.

3.4 Evaluation and inference deliverable

The notebook provides code for per-class precision/recall/F1, the 196-class confusion matrix, and five highest- and lowest-confidence predictions. Its updated evaluation and stand-alone checkpoint-loading cells did not have saved execution outputs in the reviewed Phase 3 notebook. They must still be executed and saved in Colab to satisfy the instructor's execution requirement. The checkpoint file vehicle_classifier.pth contains a ResNet50 model state dictionary, a 196-class mapping, architecture and image-size metadata. The model-loading routine reconstructs ResNet50 with the

matching output dimension and restores the saved weights. Since the checkpoint is selected using validation accuracy, the 83.87% score must not be described as the final held-out test accuracy.

4. Cross-phase discussion and lessons learned

Across the three phases, model capacity and transfer learning mattered, but so did how training data was prepared. Phase 1 demonstrated that optimization choices and initialization can make the difference between chance-level and near-perfect digit recognition. Phase 2 showed a modest improvement from augmentation and substantial performance costs when too many pretrained layers were frozen. Phase 3 combined a pretrained backbone, deterministic class mapping, augmentation, AdamW, a cosine schedule, and validation-based checkpointing for a 196-class task. Dataset difficulty differs substantially between MNIST, CIFAR10, and Stanford Cars; direct accuracy comparisons across them do not isolate the quality of any particular architecture.